

# Data visualisation of multidimensional clinical and genomic data

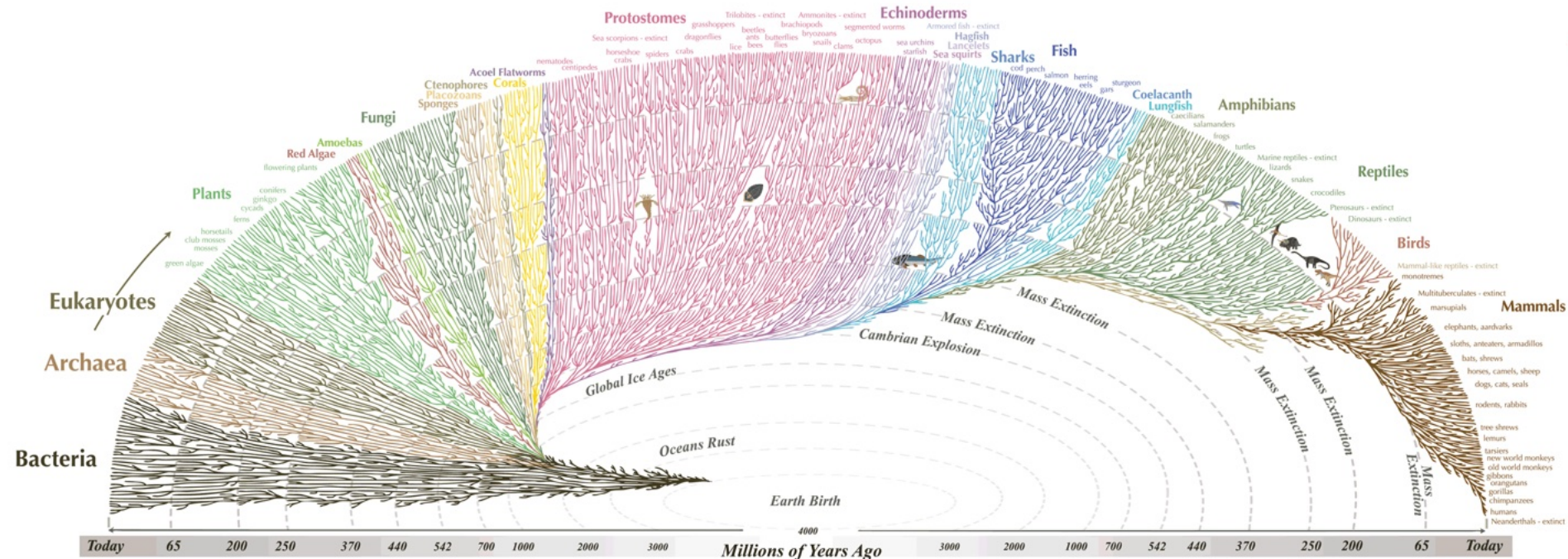
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Centre for Infectious Diseases and Microbiology – Public Health

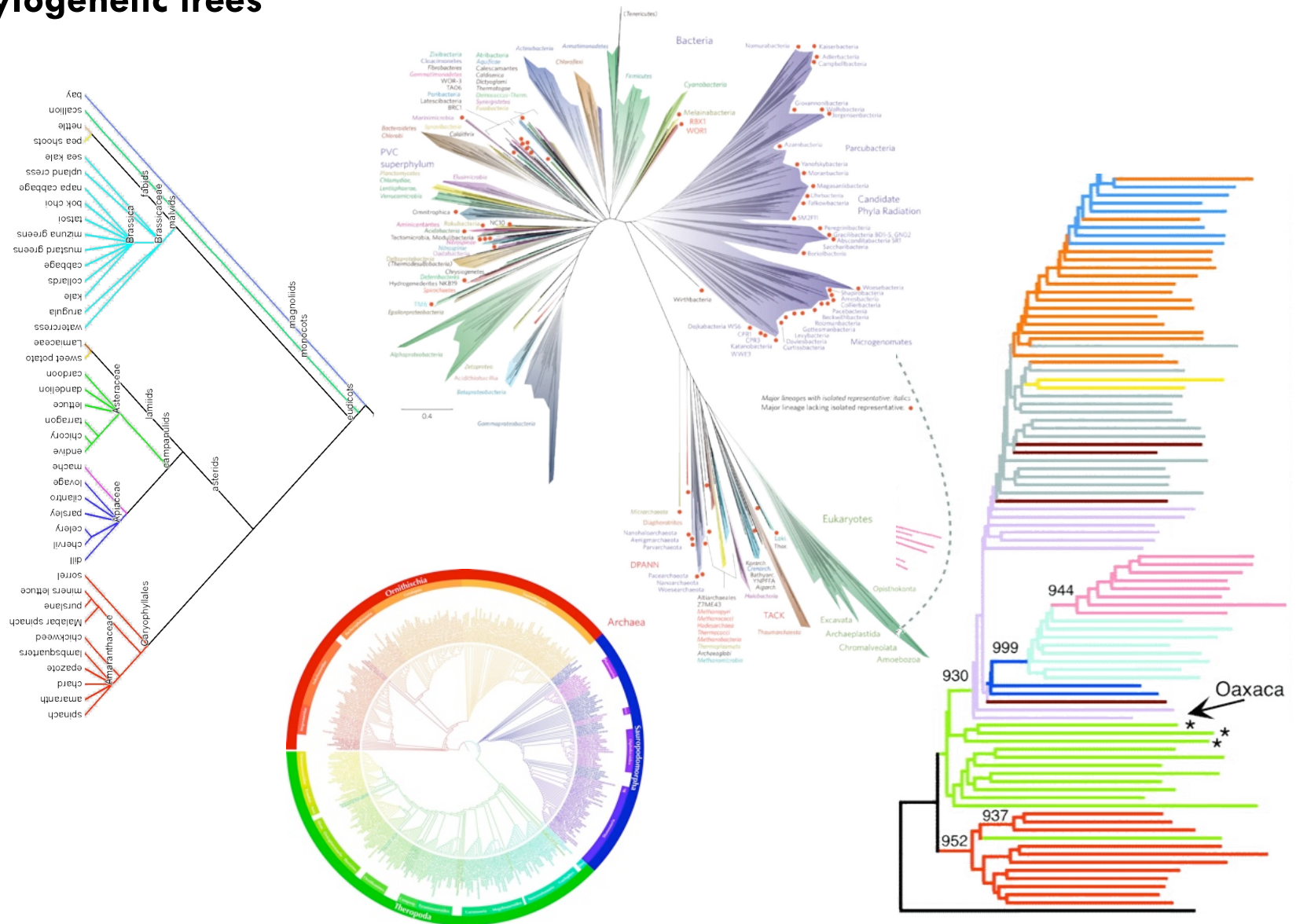
Marie Bashir Institute for Infectious Diseases and Biosecurity, University of Sydney



All the major and many of the minor living branches of life are shown on this diagram, but only a few of those that have gone extinct are shown. Example: Dinosaurs - extinct 

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evogeneao.com

# Phylogenetic trees

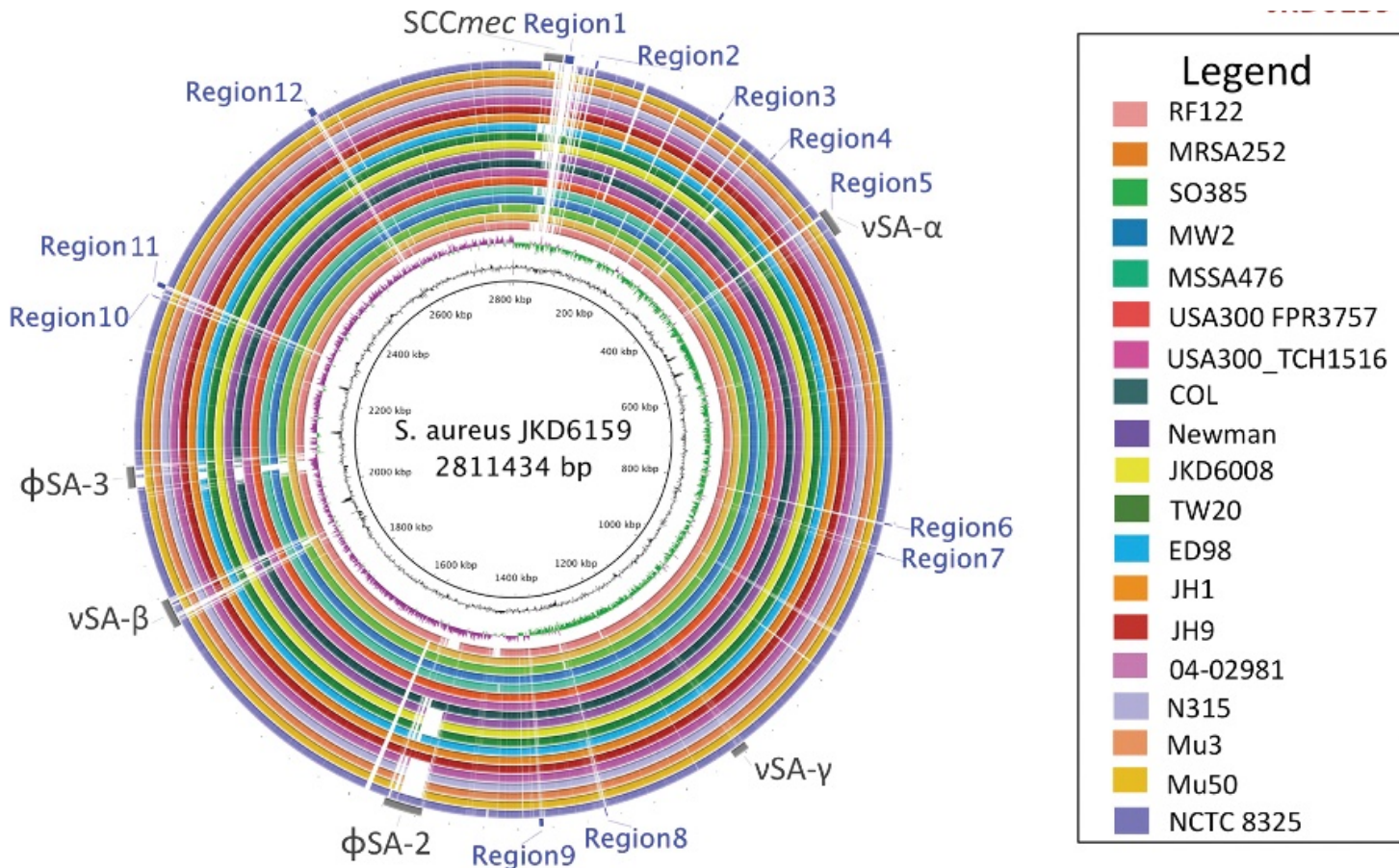


# Genome comparison

## Brig

Alikhan, et al. 2011. BLAST Ring Image Generator (BRIG), BMC Genomics

<http://brig.sourceforge.net/>



Chua, et al. 2011. CA-MRSA ST93-IV, PLoS One.

<http://dx.doi.org/10.1371/journal.pone.0025887>

# Genome comparison

## Circos

Krzywinski, et al. Genome Research (2009)

<http://www.circos.ca/>

### Close-Ups of the Genome, Species by Species by Species

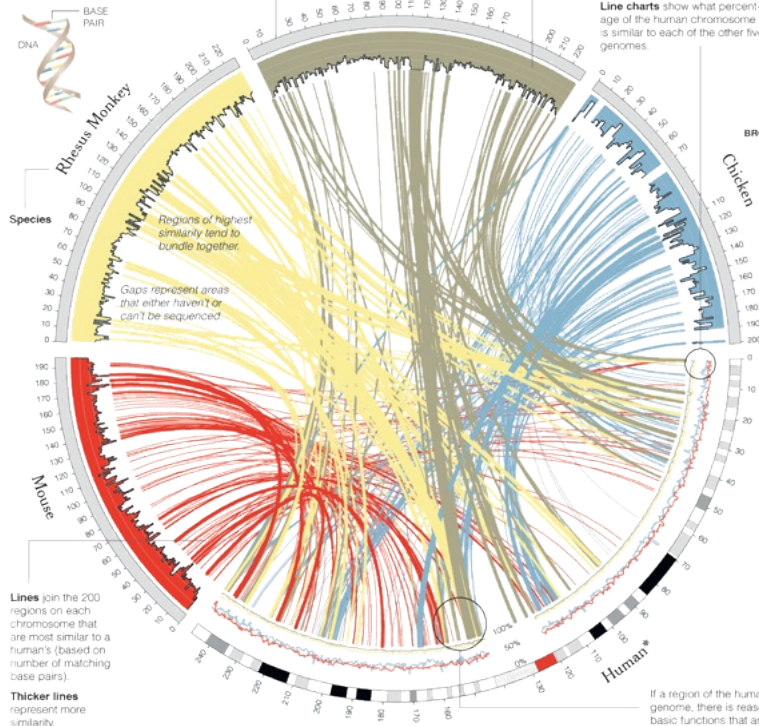
Scientists are sequencing the genomes of more than 70 organisms. The availability of these sequences has given rise to the field of comparative genomics, which seeks to answer questions about one animal's genome using information derived from another. A Canadian genomics scientist, Martin Krzywinski, has created a computer program called

Circos that aids in visualizing and comparing the data. The large diagram below illustrates the large degree of similarity between the first chromosomes of four animals to that of a human. Not surprisingly, the humans' is closest to the chimps.

DAVID CONSTANTINE

#### COMPARING CHROMOSOMES 1

Outer band represents each species' first chromosome. Numbers represent millions of base pairs on the chromosome.

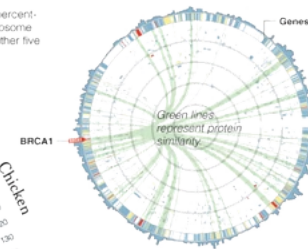


Bar charts tell how many base pairs, 0 to 1 million, match part of the human chromosome.

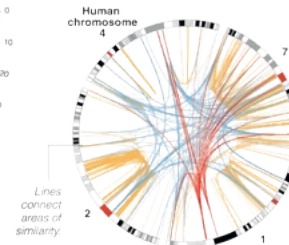
Line charts show what percentage of the human chromosome is similar to each of the other five genomes.

#### OTHER TYPES OF COMPARISONS

To download the free program or view other examples: <http://mkweb.bcgsc.ca/circos/>



The chart above shows the similarity of the BRCA1 protein, implicated in early breast cancer, to other genes on human chromosome 17.



The image above illustrates the duplication within the human genome. Here, chromosomes 1, 2, 4 and 7 are shown (arbitrarily chosen).

If a region of the human genome is very similar to a region in another's genome, there is reason to suspect that these two regions both generate basic functions that are vital to both species and do not permit variation.

Source: Martin Krzywinski, Michael Smith Genome Sciences Center, Canada

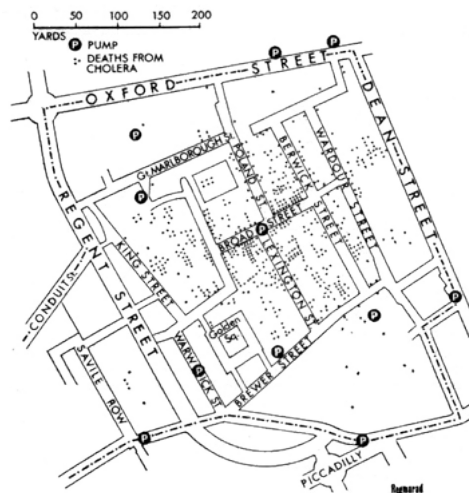
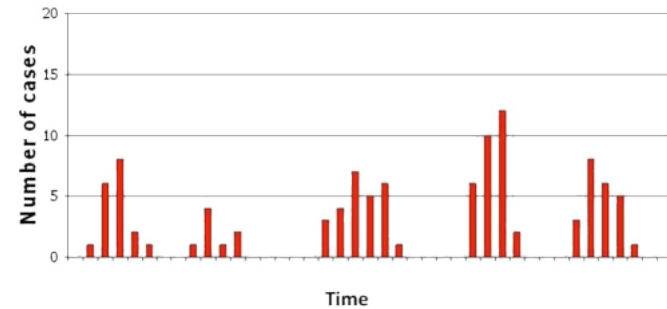
\*Length shown at 200% compared to other species. Shadings on outer band represent how chromosome looks when stained.

The New York Times

[http://www.nytimes.com/imagepages/2007/01/22/science/20070123\\_SCI\\_ILLO.html](http://www.nytimes.com/imagepages/2007/01/22/science/20070123_SCI_ILLO.html)

# Epidemiology

- Maps, heatmaps
- Line listings
- Epi curves

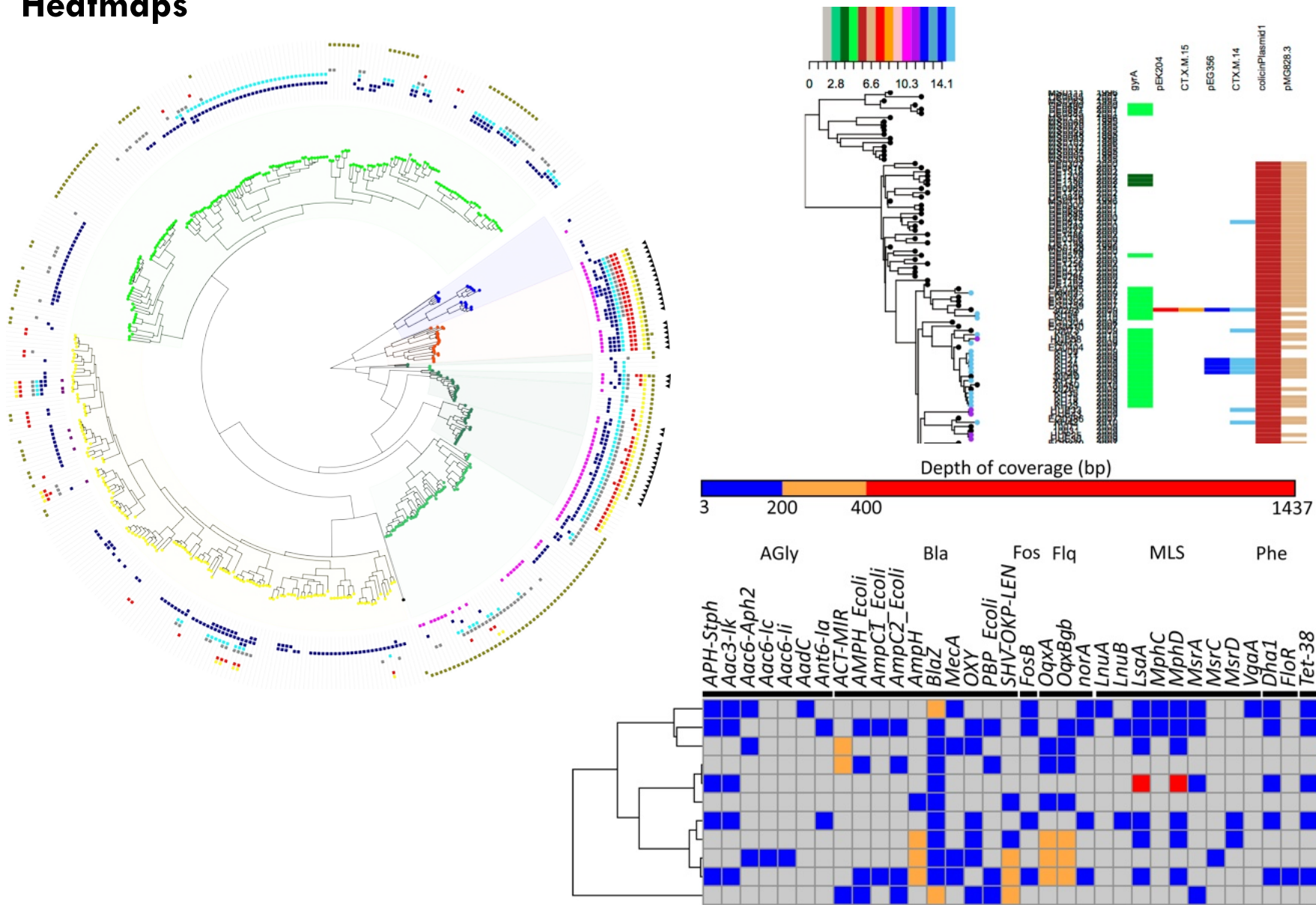


| Case# | Initials | Date of Report | Date of Onset | Physician Diagnosis | Diagnostic Signs and Symptoms |   |   |   |    |    |       | Lab        |     |     |
|-------|----------|----------------|---------------|---------------------|-------------------------------|---|---|---|----|----|-------|------------|-----|-----|
|       |          |                |               |                     | N                             | V | A | F | DU | J  | HAIGm | Other      | Age | Sex |
| 1     | JG       | 10/12          | 12/6          | Hep A               | +                             | + | + | + | +  | +  | +     | SGOT ↑     | 37  | M   |
| 2     | BC       | 10/12          | 10/5          | Hep A               | +                             | - | + | + | +  | +  | +     | Alt ↓      | 62  | F   |
| 3     | HP       | 10/13          | 10/4          | Hep A               | ±                             | - | + | + | +  | S* | +     | SGOT ↓     | 30  | F   |
| 4     | MC       | 10/15          | 10/4          | Hep A               | -                             | - | + | + | ?  | -  | +     | Hbs/Ag-    | 17  | F   |
| 5     | NG       | 10/15          | 10/9          | NA                  | -                             | - | + | - | +  | +  | NA    | NA         | 32  | F   |
| 6     | RD       | 10/15          | 10/8          | Hep A               | +                             | + | + | + | +  | +  | +     |            | 38  | M   |
| 7     | KR       | 10/16          | 10/13         | Hep A               | ±                             | - | + | + | +  | +  | +     | SGOT = 240 | 43  | M   |

S\* = Sclera; N = Nausea; V = Vomiting; A = Anorexia; F = Fever; DU = Dark urine; J = Jaundice; HAIGm = Hepatitis AIGm antibody test

# Genomic epidemiology

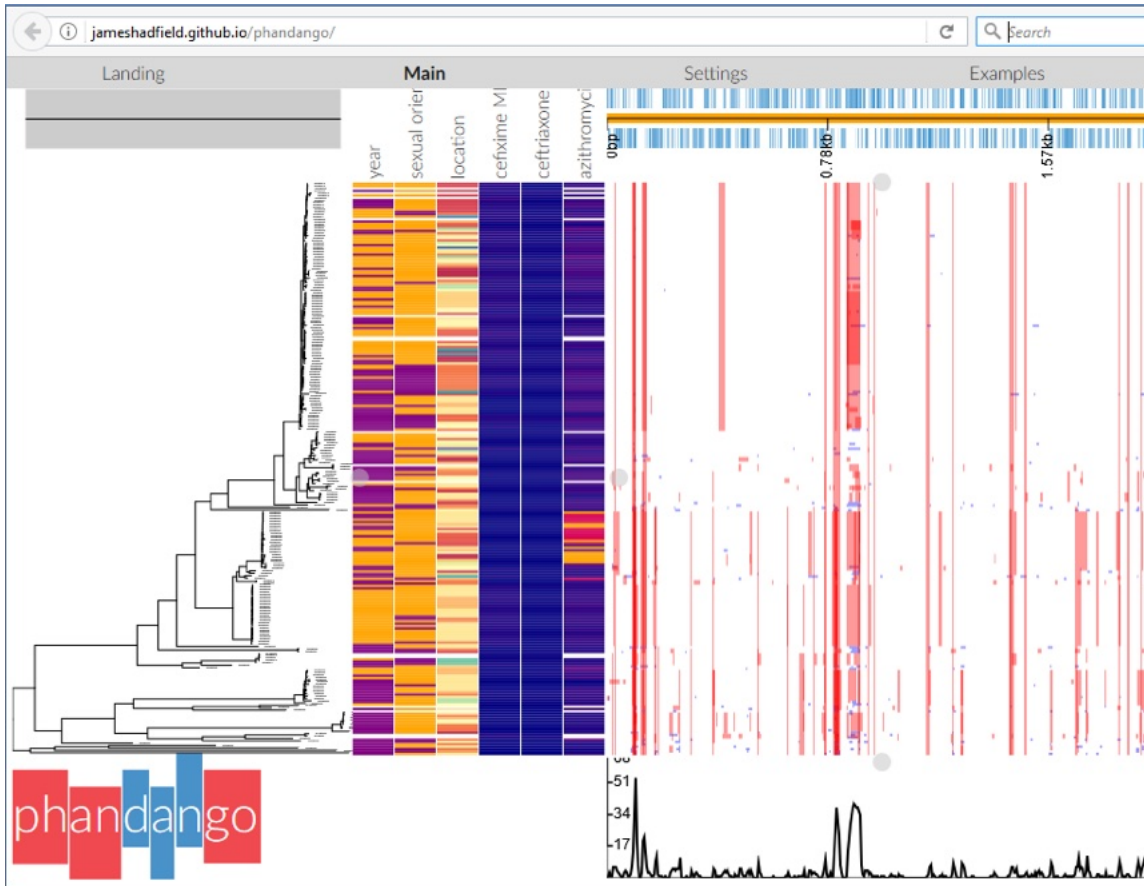
## Heatmaps



# Genomic epidemiology

## Phandango

Hadfield, Harris. <http://jameshadfield.github.io/phandango/>



*Neisseria gonorrhoeae*

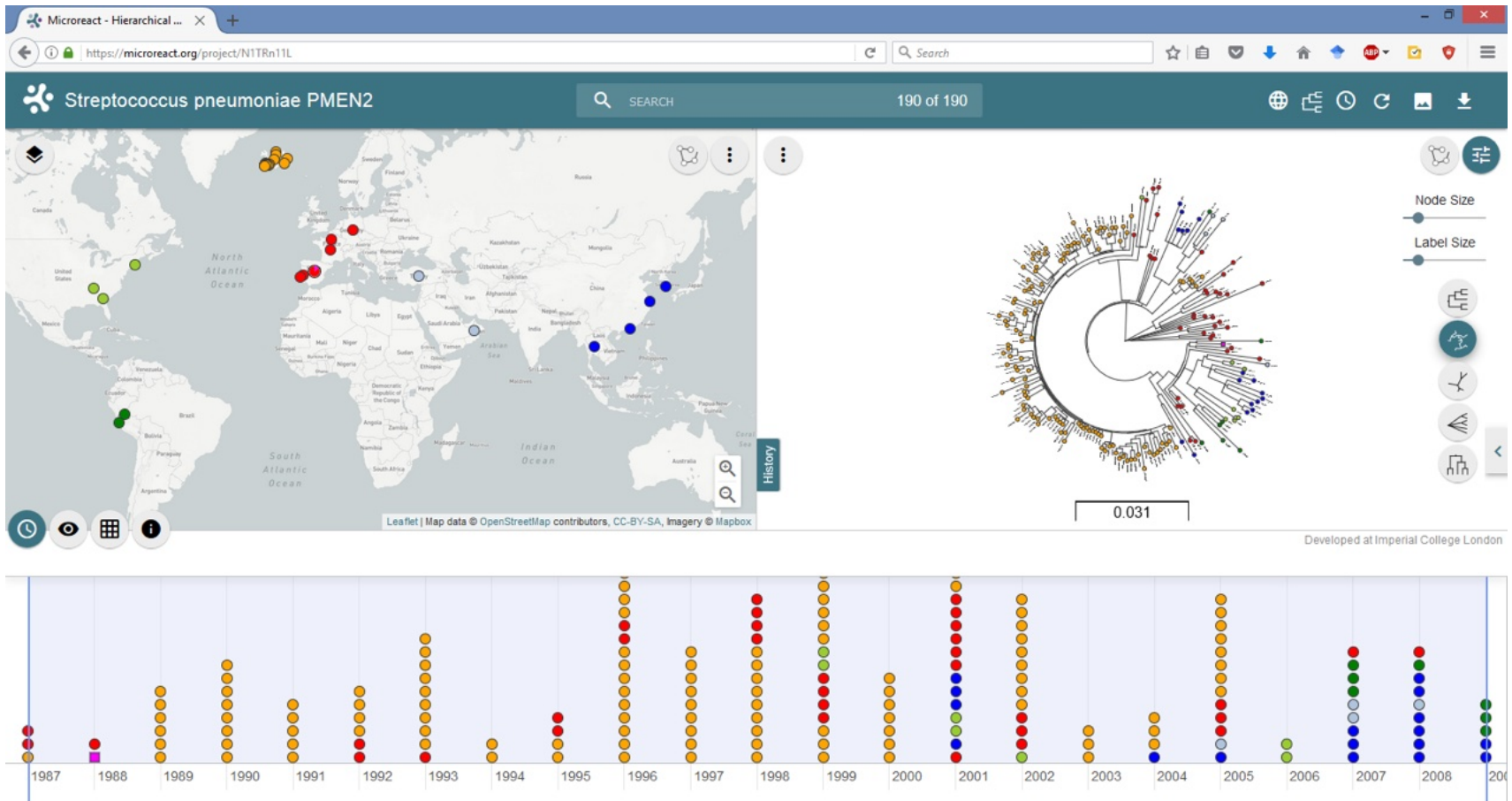
- Phylogeny
- Metadata (AB resistance)
- Recombination

# Genomic epidemiology

## Microreact

Abudahab, et al. Imperial College London, Centre for Genomic Pathogen Surveillance

<https://microreact.org/>

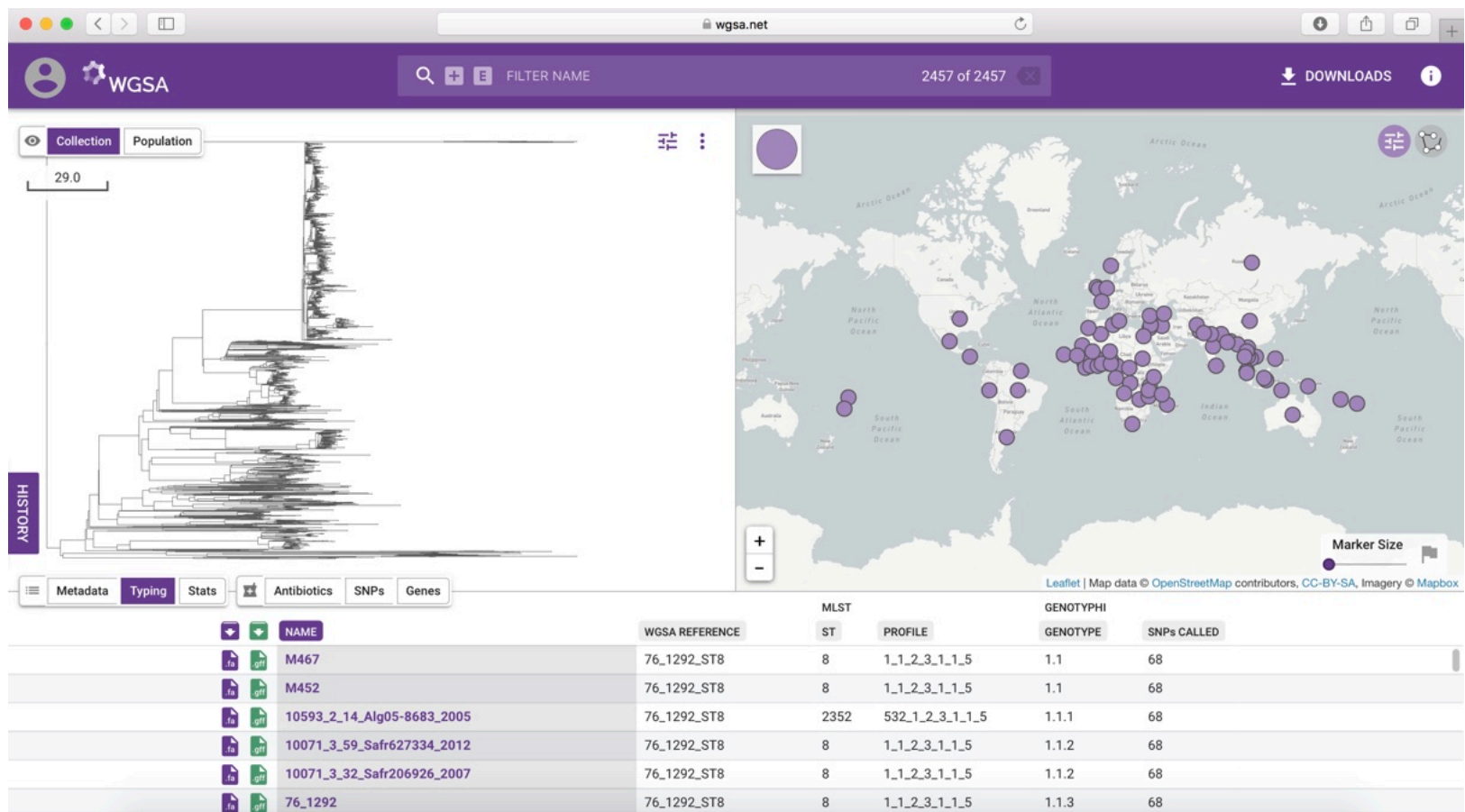


# Genomic epidemiology

## WGSA

Imperial College London, Wellcome trust

<http://www.wgsa.net/>



# Genomic epidemiology

## Integrated Rapid Infectious Disease Analysis

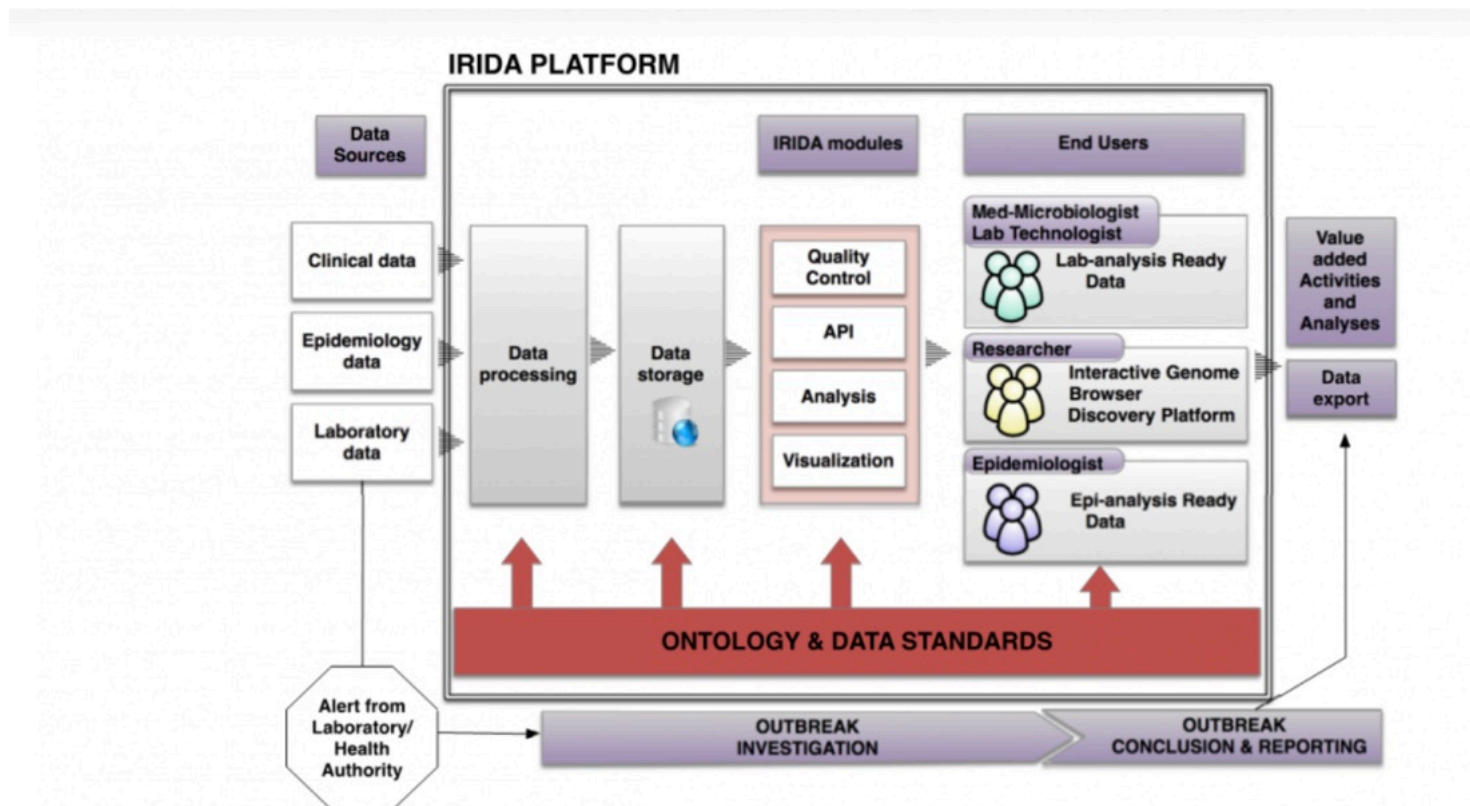
An open source, end-to-end platform for infectious disease genomic epidemiology (Canada)

<http://www.irida.ca/>



HIGHLIGHTS

DOCUM



# Genomic epidemiology

## Nullarbor

‘From reads to public health report’

Seeman, et al. <https://github.com/tseemann/nullarbor>

Browser address bar: [tseemann.github.io/nullarbor/Listeria/](https://tseemann.github.io/nullarbor/Listeria/)

### CIDM – PH Pathogen Genomics 2018

Report summary ▲

- Report summary
- Sequence data
- Species identification
- MLST
- Serotyping
- Resistome
- Assembly and annotation
- Reference genome
- Core genome
- Core SNP phylogeny
- Pairwise core SNP distances
- Core SNP density
- Pan genome
- Software versions
- About

|  |  | Date                     | Host           | Folder                                            |
|--|--|--------------------------|----------------|---------------------------------------------------|
|  |  | Tue Jun 14 23:04:34 2016 | 526080-s598782 | /mnt/seq/JOBS/BESPOKE/ASM-NGS-Conf-2015/nullarbor |

Search:

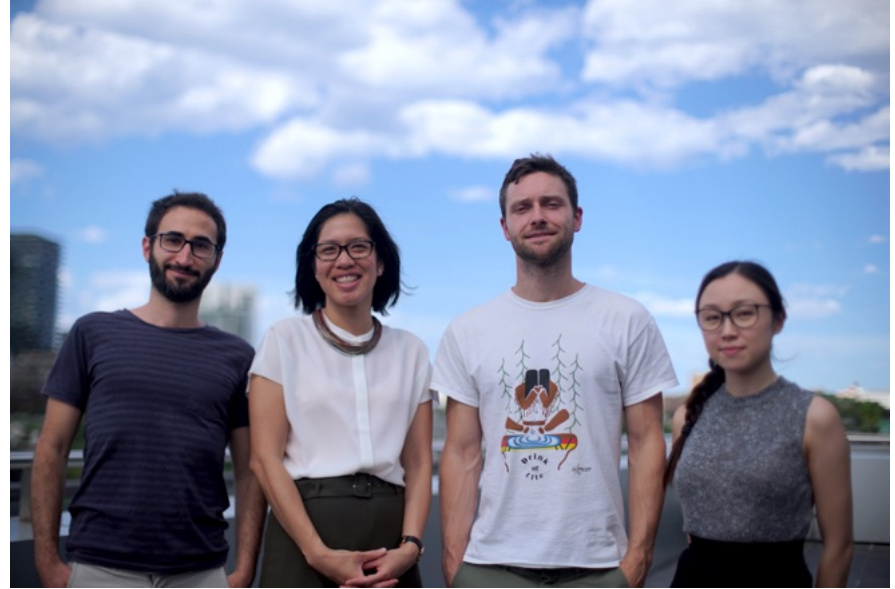
|        | Yield     | GeeCee     | MinLen | AvgLen | MaxLen | ModeLen | Phred | AvgQual | Depth | Quality |
|--------|-----------|------------|--------|--------|--------|---------|-------|---------|-------|---------|
|        | 267250486 | 38.6       | 31     | 238    | 251    | 251     | 33    | 36.3    | 84    | ✓       |
|        | 264208400 | 38.6       | 34     | 238    | 251    | 251     | 33    | 35.9    | 83    | ✓       |
| OFN-03 | 6717352   | 1513486321 | 38.6   | 30     | 225    | 251     | 33    | 35.8    | 478   | ✓       |
| OFN-04 | 5194186   | 1180263352 | 38.6   | 34     | 227    | 251     | 33    | 35.7    | 373   | ✓       |
| OFN-55 | 1701422   | 405832153  | 38.4   | 30     | 238    | 251     | 33    | 34.4    | 128   | ✓       |
| OFN-56 | 2187320   | 513849449  | 38.6   | 32     | 234    | 251     | 33    | 34.6    | 162   | ✓       |
| OFN-57 | 1790450   | 424217861  | 38.5   | 31     | 236    | 251     | 33    | 34.2    | 134   | ✓       |
| OFN-58 | 274166    | 65921121   | 38.1   | 35     | 240    | 251     | 33    | 30.1    | 20    | ✗       |
| OFN-59 | 1637386   | 385399586  | 38.5   | 34     | 235    | 251     | 33    | 34.5    | 121   | ✓       |

[tseemann.github.io/nullarbor/Listeria/#jobinfo](https://tseemann.github.io/nullarbor/Listeria/#jobinfo)

# Thank you



CIDM – PH Pathogen Genomics Team



Sydney Informatics Hub – Research Computing Team  
bioinformatics, modelling, simulation



Marie Bashir Institute  
Tackling infections, locally and globally



The Sydney Informatics Hub is a Core Research Facility. We provide support, training, and advice on research data, analyses and computing. We also collaborate on grants and projects. Talk to us about your computing infrastructure, digital tools and data governance needs. Contact [sih.info@sydney.edu.au](mailto:sih.info@sydney.edu.au) for more information.

<https://informatics.sydney.edu.au>